

Ahmed Moghazy

Data Science & ML Engineer

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Data Science graduate specializing in machine learning, applied AI, and data-driven systems. Experienced in designing and deploying end-to-end ML solutions including AutoML pipelines, predictive modeling, and retrieval-augmented generation (RAG) applications using large language models. Strong background in data analysis, feature engineering, and model evaluation, with hands-on exposure to data engineering workflows, MLOps practices, and translating complex data into actionable business insights.

Education

Cairo University — Computer Science, Bachelor of Science

Faculty of Computers & AI · Giza, Egypt · GPA 3.33 · 2021-10 to 2025-06

- Machine Learning
- Introduction to Data Science
- Mathematical Optimization
- Operating Systems
- Database Systems
- Advanced Databases
- Software Engineering
- Algorithm Design & Analysis
- Object-Oriented Programming

Experience

Advanced Computer Technology (ACT) — Software Developer (AI & Backend)

2025-10 to Present

- Architected a Multi-Agent Concierge System using LangGraph, orchestrating specialized agents with 95% intent classification accuracy.
- Engineered a Hierarchical RAG Pipeline in PGVector, reducing hallucinations by 30% on long-form queries.
- Integrated LLM Tool-Access with Oracle Opera PMS via REST APIs, reducing manual front-desk workload by 40%.
- Containerized the entire multi-agent stack using Docker, streamlining CI/CD pipeline.

Digital Egypt Pioneers Initiative (DEPI) — Machine Learning Engineer Trainee

2024-10 to 2025-04

- Completed the Microsoft ML Engineer track, focusing on scalable production-ready ML solutions.
- Built foundations in statistics, linear algebra, Python, classical ML, deep learning, NLP, and computer vision.
- Applied Azure AI Fundamentals and Azure AI Engineer Associate concepts with MLflow and Hugging Face.

Orange Egypt — Data Engineer Intern

2024-08 to 2024-09

- Automated and optimized ETL workflows for company voucher data using Apache NiFi and dbt models.
- Explored pipeline orchestration and scheduling using Apache Airflow.

ValU — Data Science Intern

2024-07 to 2024-07

- Analyzed company growth metrics and customer behavior for data-driven strategic insights.
- Designed interactive Power BI dashboards visualizing KPIs for executive decision-making.
- Built ML models (Random Forest, XGBoost) for customer lifetime value prediction, achieving AUC of 0.70.
- Contributed to a customer-facing chatbot using NER for accurate offer retrieval.
- Optimized LLM prompts for intent extraction, reducing inference costs by ~30%.

Advanced Computer Technology (ACT) — Data Science Intern

2023-08 to 2023-09

- Conducted EDA on large datasets and presented insights to stakeholders.
- Automated web scraping pipelines using BeautifulSoup.

Projects

AI-Powered Tech News Aggregator — n8n, LLMs

2026-01 to 2026-01

- Built an automated news aggregation pipeline using n8n to ingest RSS feeds on a scheduled basis.
- Implemented LLM-based classification and ranking to categorize articles and score importance.
- Designed workflows to store results in Google Sheets and deliver curated HTML email summaries.

AutoML Pipeline — scikit-learn

2024-10 to 2025-06

- Engineered a modular AutoML pipeline with 4 stages: preprocessing, feature selection, model selection, and HPO.
- Synthesized AutoML foundations into the system architecture and experiment plan.
- Benchmarked Grid Search, Random Search, and Bayesian Optimization for HPO on tabular ML tasks.

RAG Chatbot — LangChain + FAISS + Ollama

2024-12 to 2025-01

- Built a retrieval-augmented chatbot with LangChain + Ollama Gemma-3 (12B) and Gradio UI.
- Crawled 30 LangChain doc pages; chunked via RecursiveCharacterTextSplitter.
- Embedded with all-MiniLM-L6-v2, indexed in FAISS; used MultiQueryRetriever + ContextualCompressionRetriever.

ExpenSum — Smart Expense Tracker — React + Spring Boot + JWT + LLM

2025-04 to 2025-05

- Developed a full-stack expense tracker: React frontend + Spring Boot backend secured with JWT.
- Integrated Mistral (via Ollama) to convert natural-language inputs into structured expense entries.

Star-Schema Data Warehouse — SQL, ETL

2023-12 to 2024-01

- Designed a star schema (fact/dimension) for a recommendation dataset.
- Automated daily CSV loads via scheduled SQL jobs (ETL).

Certifications

Deep Learning Specialization — DeepLearning.AI

2024-10 to 2024-12

ICT Global Competition — AI Section Lead (Cloud Track) — Huawei

2024-11 to 2025-05

- Won the Second Prize Globally in the Huawei ICT Global Competition (Cloud Track).
- Awarded the Grand Prize in the North Africa Region, representing Egypt in the global finals.
- Led the AI Technologies section, covering Machine Learning & Deep Learning.

Skills

Programming

Python (Advanced), SQL, C++, JavaScript, HTML/CSS

Machine Learning & AI

Scikit-learn, TensorFlow/Keras, Classical ML, Deep Learning, NLP, Computer Vision, Feature Engineering, Model Evaluation, HPO

LLMs & Generative AI

LangChain, RAG, Prompt Engineering, Hugging Face, Ollama, FAISS, Vector Databases

Data Analysis & Visualization

Pandas, NumPy, Matplotlib, Seaborn, Power BI, EDA

Data Engineering & MLOps

Apache Airflow, Apache NiFi, dbt, ETL Pipelines, MLflow, HDFS

Databases

PostgreSQL, MySQL, Star Schema Design, Data Warehousing

Cloud & Tools

Azure AI Fundamentals, Huawei Cloud, Git/GitHub, Docker, Linux